



SUBHASH PALLETI

subhash.palleti2025@gmail.com | +1 816-572-0999

PROFESSIONAL SUMMARY:

- Certified Data Engineer with over 6+ years of experience in designing and implementing data pipelines, managing data warehouses, and ensuring data integrity in both cloud and on-premises environments.
- Experience in Extraction, Transformation and Loading (ETL) data from various sources into Data Warehouses, as well as data processing like collecting, aggregating and moving data from various sources using Apache Flume, Kafka, PowerBI and Microsoft SSIS.
- Experienced Cloud Computing Specialist with over 5 years of expertise in deploying, managing, and optimizing AWS services (EC2, RDS, S3, VPC, CloudFormation).
- Skilled in containerization technologies like Docker and Kubernetes to enhance scalability and portability of applications.
- Adept at database administration for MySQL, MSSQL, PostgreSQL, DynamoDB, MongoDB, and Elasticsearch, ensuring optimal performance and availability.
- Imported the data from various formats like JSON, Text, CSV, AVRO, and Parquet to HDFS cluster with compressed for optimization.
- Comprehensive experience working in regulated environments (HIPAA, PCI, ITAR), ensuring compliance and security standards.
- Implemented in Unix, Amazon DynamoDB, AWS EC2, AWS EMR, AWS Glue, AWS Lambda, SQL, Spark, and Python to ensure robust data infrastructure.
- Utilized Apache Spark with Scala to develop and execute Big Data Analytics and Machine learning applications, executed machine learning use cases under Spark ML and MLlib.
- Managed and optimized AWS Redshift data warehouses, designing schemas, optimizing query performance using distribution and sort keys, and ensuring efficient data storage with compression techniques.
- Integrated Apache Kafka for handling high-throughput, low-latency real-time data ingestion and processing in a highly scalable architecture.
- Expert in designing Parallel jobs using various stages like Join, Merge, Lookup, remove duplicates, Filter, Dataset, Lookup file set, Complex flat file, Modify, Aggregator, and XML.
- Developed and maintained real-time data processing pipelines using AWS Kinesis, Lambda, and PySpark for streamlining data from multiple sources to Redshift.
- Developed and maintained graph-based data models using Neo4j, enabling efficient relationship mapping and data traversal across complex datasets.
- Translated business relationships into graph structures, utilizing Cypher queries to perform advanced pattern matching and analytics.
- Proficient in networking and security, implementing robust solutions for data protection and access control.
- Expertise in configuration management using frameworks like Puppet, Chef, and Ansible to automate infrastructure deployment.
- Proven ability in designing and maintaining CI/CD pipelines, integrating automated testing frameworks, and utilizing DevOps tools (XLR, Jenkins, Git, Maven).

- Exceptional communication and analytical skills, including effective presentation abilities to convey complex technical concepts.
- Familiar with Agile methodologies and Scrum ceremonies, ensuring efficient project delivery and team collaboration.
- Experience in Converting existing AWS Infrastructure to Server less architecture (AWS Lambda), deploying via Terraform and AWS Cloud Formation templates.
- Hands-on experience in developing and deploying enterprise-based applications using major Hadoop ecosystem components like MapReduce, YARN, Hive, HBase, Flume, Sqoop, Spark MLlib, Spark GraphX, Spark SQL, Kafka.
- Good understanding of Spark Architecture with Databricks, Structured Streaming. Setting Up AWS and Microsoft Azure with Databricks, Databricks Workspace for Business Analytics, Manage Clusters In Databricks.
- Developed Spark Applications that can handle data from various RDBMS (MySQL, Oracle Database) and Streaming sources. Proficient experience in using Python lists, panda's, tuples, sets, dictionaries, loops, classes, object-oriented programming, JSON, and logging concepts.
- Improved CI/CD processes for data pipeline deployment and enhanced DevOps practices.
- Built Tabular Models in AAS for real-time OLAP and advanced BI solutions.
- Integrated AAS with Power BI, enabling interactive data visualization and reporting.
- Provided on-call production support, troubleshooting failures, and optimizing pipeline performance.
- Configured real-time data exploration using ADX for log and telemetry analysis.
- Created KQL-based dashboards in ADX to monitor business metrics and application performance.
- Practically Designed and Implemented Control Flow, Data Flow, Data Movement Activities using ADF.
- Ensured adherence to PII encryption and masking standards for data security and compliance.
- Actively participated in Agile ceremonies and contributed to the Engineering Community of Practice.
- Demonstrated experience on PySpark API transformations, actions and PySpark optimizations.
- Expertise in writing UDF using PySpark, Python, T-SQL to drive business insights.
- Experienced in writing complex SQL, T- SQL scripts, stored procedures, triggers, clustered and non-clustered indexes, views, strings, date functions, analytical functions, CTE and optimizations.
- Designed and implemented message-driven architectures using Google Pub/Sub for real-time data streaming and event driven workflows.
- Strong analytical and troubleshooting skills for Azure Compute environments, ensuring quick resolution of issues.
- Excellent team collaboration and communication skills, fostering innovation in cloud-based solutions.
- Strong knowledge of data compliance regulations (GDPR, CCPA, HIPAA, SOC2).
- Proficient experience in UNIX shell scripting and automating the functionalities.
- Strong experience with Azure DevOps, Git for continuous integration and deployment CICD pipelines.
- Highly experienced in SDLC and Agile implementation methodologies.

SKILLS:

Languages	SQL, T-SQL, Python, Java, C, C++, C# /.NET, R, JavaScript.
Operating Systems	Unix, Linux, macOS.
Cloud Platforms	AWS, Azure, GCP.
Big data Tools	ADF, Azure Databricks, Azure Synapse Analytics, Azure SQL, Azure Logic Apps, HDFS, Hive, YARN, Spark, PySpark, Sqoop, Apache Airflow, Control-M, Hadoop, Hadoop Ecosystem Map Reduce, Nifi, HBase, Hive, Pig, Oozie.

Data Streaming Tools	Apache Kafka, Apache Spark Streaming, Amazon Kinesis, Google Cloud Dataflow, Azure Event Hubs.
Data Visualization Tools	Power BI, Tableau, Google Data Studio, OBIEE.
CI/CD Build Tools & Containerization Tools	Azure DevOps, Team city, Git, Jenkins, Kubernetes, Docker, ANT, Maven, Hudson, Bamboo.
Data Modelling	Star and Snowflake Schema, entity-relationship (ER), object-oriented and graph models, Erwin Data Modeler, ER Studio V17.
Data Warehousing	Data Vault, Data Marts, Slowly Changing Dimensions SCD-1,2,3.
Data Storage & Management Tools	Azure Blobs, Amazon S3, Apache HDFS, Azure Data Lake Storage, Delta Lake.
Project Management Tools	JIRA, Confluence, Azure Devops, GIT, Bitbucket, Terraform.
IDE & Build Tools	PyCharm, Jupyter Notebook, Eclipse, Microsoft Excel, SQL Server Management Studio (SSMS).
Data Governance Tools	Azure Key Vault, Azure Active Directory, Azure purview, Unity Catalog.
Data structures	Arrays, Linked Lists, Stacks, Queues, Sorting, Searching.
Data Bases	Oracle, SQL Server, MySQL, Snowflake, PostgreSQL, DynamoDB, MongoDB, Netezza, COSMOS DB
Data Science	Machine Learning, Deep Learning, NLP, PyTorch, TensorFlow
CI/CD Tools	JENKINS, HUDSON, Anthill Pro, Nexus, Ansible, Docker Swarm, Kubernetes, Power Shell Script, Terraform, Azure Devops.

CERTIFICATIONS:

- **MICROSOFT CERTIFIED AZURE DATA ENGINEER- DP - 203.**
- **AZURE DATABRICKS.**
- **SNOWFLAKE.**
- **INTRODUCTION TO PYTHON.**
- **DATA SCIENCE.**

EDUCATION:

- **Masters | Computer Science | University of Central Missouri, MO | 3.4/ 4.0 |**

EXPERIENCE:

Data Engineer, JPMC, Atlanta, GA

Aug 2023 – Present

- Migrated on premises data mart data to Azure data lake gen2/Azure synapse using azure data factory and Azure Data Bricks.
- Based on new requirements, Analyzed the existing data model and Built dimensions and facts and loaded those using Azure Databricks.
- Implemented SCD type1, SCDType2 logic using azure data bricks for Delta tables.
- Developed pyspark-streaming applications and loaded those using techniques like SCD type1, SCD type2.
- Analyze performance of pyspark job and improved them by using performance techniques like, partitioning, partitioning pruning.
- Functional knowledge of enterprise ETL platforms, including IBM DataStage, Informatica, Pentaho, and Ab Initio. Created dynamic code to pull incremental data from on premises to Azure Data Lake gen2.
- Mounted Azure Data Lake Gen2 to Azure data bricks.
- Created key vault and secret keys for storage accounts and loaded the data using Data Bricks.
- Created ADF pipelines to load the data from AWS S3 to Azure data lake gen2.

- Created azure devops CI/CD pipelines to create Pull request and to migrate in production.
- Created Store procedures in Azure SQL DB to implement Audit log mechanism.
- Deployed and orchestrated containerized applications using Azure Kubernetes Service (AKS) and Azure Container Instances (ACI).
- Experience in data modeling and working with relational databases, NoSQL, and other data structures.
- Hands-on experience with data modeling tools such as Erwin, IBM Infosphere Data Architect, or similar tools.
- Solid understanding of various data models, including conceptual, relational, dimensional, and NoSQL.
- Experience with ETL processes and data warehousing concepts, tools, and techniques.
- Ability to gather and interpret business requirements and translate them into actionable data models and diagrams.
- Knowledge of key performance indicators (KPIs) and data usage patterns to design effective and meaningful data models.
- Strong analytical and problem-solving skills with the ability to translate complex business needs into structured data models.
- Experience working closely with business stakeholders and cross-functional teams to ensure alignment of business and data requirements.
- Excellent written and verbal communication skills with the ability to present complex technical concepts to non-technical stakeholders.
- Extensive use of DAX (Data Analysis Expressions) functions for the reports and for the tabular models.
- Created Power BI reports by using Joins in multiple tables from multiples database using complex SQL queries.
- Skilled in modern cloud development using Terraform, Azure CLI, and PowerShell.
- Experience working in multi-developer environments and using Azure DevOps or GitLab for versioning and pipeline management.
- Apply the Business rules on the data coming from multiple operating companies. Maintain the business rules in a table for visibility to all required processes to develop Power BI reports & effective dashboards.
- Load the processed data into final viewpoint centralized DWH for further Power BI reporting.
- Worked on Key Performance Indicators (KPIs), design of star schema and snowflake schema in Analysis Services (SSAS).
- Designed and implemented a fully operational production grade large scale data solution.
- Performed data wrangling to clean, transform and reshape the data utilizing panda's library. Analyzed data using SQL, Scala, Python, Apache Spark and presented analytical reports to management and technical teams.
- Strong programming skills in Java, including expertise in data structures, algorithms, and object-oriented design principles.
- Developed stored procedures/views in Snowflake and use in Talend for loading Dimensions and Facts.
- Building a Scala and spark based configurable framework to connect common Data sources like MYSQL, Oracle, Postgres, SQL Server, and Salesforce and load it in big query.
- Create reports for the BI team using Sqoop to export data into HDFS and Hive, Configure and install Hadoop and Hadoop ecosystems (Hive/Pig/ HBase/ Sqoop).
- Proficiency in Kubernetes and Docker for deploying containerized applications.
- Designed and implemented compute solutions using Azure Virtual Machines, Azure Kubernetes Service (AKS), and Azure Functions to optimize system performance and scalability.
- Developed and deployed applications leveraging Azure Compute services, ensuring cost-effectiveness and robust infrastructure.
- Utilized Kusto Query Language (KQL) to analyze and process log data for real-time monitoring dashboards.
- Developed time-series analytics in Kusto, supporting predictive maintenance solutions.
- Automated deployment pipelines for Azure Compute services, utilizing tools like Azure DevOps to streamline workflows and reduce downtime.
- Designed and implemented database-first and code-first models using Entity Framework in .NET applications.

- Collaborated in Agile/Scrum environments, actively participating in sprint planning, daily stand-ups, and retrospectives.
- Engineered large-scale data processing solutions using COSMOS, enabling real-time analytics across billions of rows.
- Ensured optimal performance of COSMOS jobs by fine-tuning partitions and resource utilization.
- Connected and interacted with diverse DBMSs like SQL Server, DB2, and Oracle to enable seamless data access and migration.
- Integrated Big Query, Cloud Storage, and Cloud Pub/Sub for unified data processing and reporting workflows.

Data Engineer, Tata Consultancy Services, India

Aug 2020 – Jun 2023

- Configured Azure Event Hubs to ingest data from various retail sources such as POS systems, online transactions, and customer interaction platforms, ensuring reliable and scalable data flow.
- Designed and configured Azure Stream Analytics jobs to process incoming data in real-time.
- Performed Data Profiling and Business data validation rules.
- Ingested bulk data from different sources using Azure Data Factory (ADF), automating data workflows and ensuring timely data availability.
- Integrated pre-trained models from Azure Cognitive Services into data workflows to streamline natural language processing and image recognition tasks.
- Designed and implemented custom deep learning models for predictive analytics, leveraging frameworks like TensorFlow and PyTorch.
- Developed real-time streaming pipelines for inference using Azure Stream Analytics and deployed trained models via Azure Kubernetes Service (AKS).
- Processed real-time event data and generated actionable insights by deploying AI models in production systems.
- Developed and deployed machine learning models using Azure Machine Learning and automated model retraining pipelines for improved accuracy over time.
- Orchestrated model versioning and deployment in production environments using MLOps best practices, ensuring robust and reproducible workflows.
- Integrated Synapse Analytics and Azure Data Lake for seamless model deployment with advanced data integration.
- Enhanced data quality and feature engineering for ML models by applying statistical analysis and domain-specific transformations in Python.
- Automated feature selection and extraction processes, significantly reducing time spent on data preparation for ML models.
- Led cloud migration initiatives, transitioning on-premises data marts to Azure Data Lake Gen2 and Synapse, ensuring minimal downtime and data consistency.
- Migrated enterprise data systems to cloud platforms (Azure/GCP), integrating with tools like Databricks and Snowflake for scalable data processing.
- Designed and implemented data warehouse architectures using star and snowflake schemas to support multi-dimensional data analysis.
- Applied dimensional modeling techniques to create scalable and optimized schemas for analytics and reporting use cases.
- Implemented Slowly Changing Dimensions (SCD Type 1 & 2) using PySpark and ADF to track historical data effectively.
- Developed Spark applications using PySpark and SQL Databricks notebooks to perform complex data transformations and load cleansed data into Azure SQL DB, improving data quality and readiness for analytics.
- Identified Facts and Dimensions to implement a star schema, enhancing data organization and facilitating efficient querying. Implemented data models in Azure SQL Data Warehouse (DWH), creating tables and schema optimized for analytics and reporting.
- Optimized the performance of Azure SQL by managing indexes, partitioning tables, managed and external tables and optimizing queries, resulting in faster data retrieval and processing times.

- Ensured data integrity and consistency by implementing data validation and data quality checks at various stages of the data pipeline.
- Designed and optimized secure API gateways for seamless integration with Pi's partner merchants, enabling real-time tracking of cashback offers and user redemptions.
- Monitored and troubleshooted data pipelines, ensuring high availability and reliability of data processing workflows.
- Collaborated with cross-functional teams, including data scientists, analysts, business intelligence, and business stakeholders, to understand requirements and deliver data solutions that meet business needs.
- Provided technical guidance and mentorship to junior data engineers, fostering a culture of continuous learning and improvement. Implemented data security best practices, including encryption and access controls, to protect sensitive customer and business data.
- Ensured adherence to security protocols and compliance standards across AWS environments, maintaining a secure cloud infrastructure.
- Established CI/CD pipelines using AWS services like Code Commit, Code Build, Code Deploy, and Code Pipeline. Implemented best practices for CI/CD, including automated testing and deployment strategies.

Big Data Engineer, Hytronics Enterprise, India

Jun 2018 – July 2020

- Utilized Sqoop for importing and exporting data between relational databases and HDFS.
- Thoroughly plan the ETL process, understanding source data and defining transformation rules.
- Integrated Terraform with version control systems like Git for collaborative development and code management, enabling versioning and rollback capabilities.
- Implemented Azure DevOps pipelines for continuous integration and continuous deployment (CI/CD), ensuring automated and streamlined application delivery.
- Design the ETL architecture considering performance, scalability, and maintainability.
- Developed custom Python / PySpark scripts for data ingestion and processing.
- Skilled in database performance tuning and real-time service call optimization.
- Leveraged Pandas and NumPy for efficient preprocessing and statistical analysis of survey responses, enabling data-driven insights for employee feedback trends.
- Designed a dynamic visualization dashboard using Plotly and Power BI, allowing admins to analyze survey results, track engagement trends, and generate actionable insights.
- Created hive queries for data analysis and wrote customized functions to meet business requirements.
- Developed ETL data pipelines from Sources Systems to AWS services using a combination of Databricks, Spark SQL, Redshift AWS Data and Delta Lake Analytics.
- Designed and implemented database solutions in Snowflake Data Warehouse.
- Created pipeline jobs, Scheduling triggers, PySpark, and mapping data flows using AWS Cloud composer with Airflow.
- Worked on ingesting data from RDBMS sources like Oracle Server, and Teradata into HDFS using Sqoop.
- Creating Components using Mahout Algorithms, Written Hive Queries to visualize the data for the prediction of algorithm.
- Creating charts for algorithm results using hive queries using BDE tool.